1560–64

Ambroise Paré developed new surgical techniques while working as a battlefield surgeon, and demonstrated his ideas to his students.

GABRIELE FALLOPPIO (1523–62)

Born in Modena, Italy, Falloppio studied medicine in Ferrara, and went on to teach anatomy and surgery at the Universities of Ferrara and Padua. He also served as superintendent of the botanical garden at Padua.

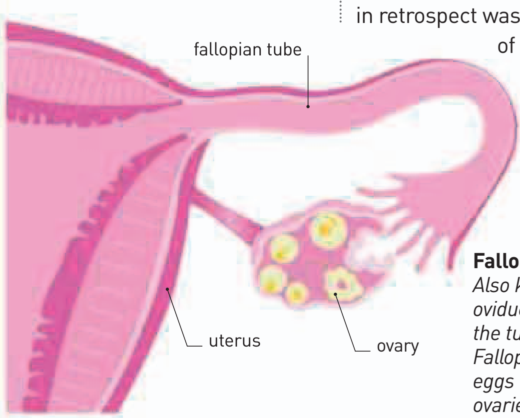
He is credited with important discoveries in the anatomy of the human head and reproductive systems. He died, aged only 39, in Padua.

his treatise ***De Re Anatomica*** (*On Things Anatomical*). Colombo's practical background in surgery led to a sometimes acrimonious rivalry with his more academic contemporary Andreas Vesalius. However, he is credited with **advances in anatomy**, including work on **pulmonary circulation**.

> “IT IS GOOD FOR NOTHING BUT TO **CHOKE A MAN**, AND FILL HIM FULL OF **SMOKE AND EMBERS.**”
>
> **Ben Jonson, English playwright,** in *Every Man in His Humor*, 1598

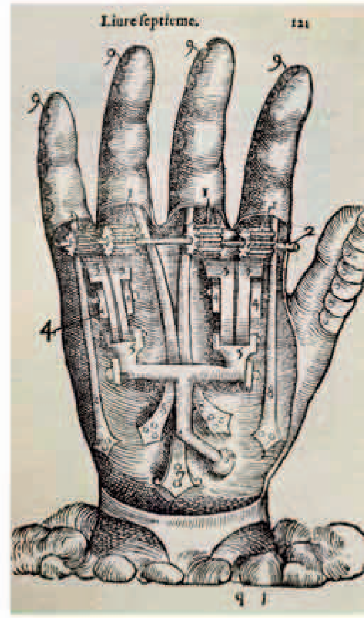
ITALIAN POLYMATH PLAYWRIGHT GIAMBATTISTA DELLA PORTA (*c.*1535–1615), particularly interested in the sciences, founded a group of like-minded thinkers in Naples, nicknamed the Otiosi, or men of leisure, who met to "uncover the secrets of nature." Their more formal title was the **Academia Secretorum Naturae** (*Academy of Secrets of Nature*), considered to be the **first scientific society**. Its membership was open to anyone who could demonstrate having made a new discovery in one of the natural sciences, and meetings were held at della Porta's home until, thanks to his interest in occult philosophy, Pope Paul V ordered the Academy to disband in 1578. Della Porta went on to encourage the founding of another society, the **Accademia dei Lincei** (Academy of Lynxes) in 1603.

Gabriele Falloppio, who had succeeded Realdo Colombo as chair of anatomy and surgery at Padua University in 1551, published his major work, ***Observationes Anatomicae*** **(*Anatomical Observations*)**, in 1561 at the height of what in retrospect was a golden age of **anatomical discovery**. Sometimes known by the Latinized form of his name, Fallopius, he made valuable contributions to the **study of the ears, eyes, and nose, as well as human reproduction and sexuality**. A nerve canal in the face (the *aquaeductus Fallopii*) and the Fallopian tube connecting the ovaries with the uterus are named after him. Falloppio was a respected physician and skillful surgeon as well as an anatomist. He wrote a number of treatises on surgery, medication, and treatments of various kinds, although only one, *Anatomy*, was published in his lifetime. His work complemented the writings of his compatriots Vesalius and Colombo, and often quietly corrected their misconceptions.



Fallopian tube
Also known as the oviducts or salpinges, the tubes named after Falloppio allow the eggs to pass from the ovaries to the uterus.

Meanwhile, Frenchman **Ambroise Paré** (1510–90) wrote one of the **first manuals of modern surgery**. His ***Treatise***, written in French rather than Latin, was based on his experience as a surgeon on the battlefield. As well as describing surgical procedures, some of his own invention, Paré's book proposed the idea of surgery as a **restorative procedure**, which should involve minimum suffering. It stated that pain relief, healing, and even compassion were essential to successful surgery. This came from his experience of treating amputated limbs with balms and ointments rather than cauterizing the wounds with boiling oil, which often harmed the very tissues the surgeon was attempting to mend. Paré's innovative scientific approach based on **empirical observation** did much to improve the status of the "barber-surgeon," which was previously considered inferior to the medical physician.



Prosthetic hand by Paré
Paré designed sophisticated prostheses to replace missing limbs, such as the hand shown in this 1585 drawing.

1559 Jean Nicot introduces **tobacco** to France

1559 Realdo Colombo publishes his treatise ***De Re Anatomica***

1560 The **first scientific society** is founded in Naples by Giambattista della Porta

1561 Gabriele Falloppio describes the human **ovaries, the uterus, and the tubes** connecting them

1564 Ambroise Paré publishes his ***Treatise*** on surgery